

Student Guide — senior / module-02- panchabhuta

IKS Curriculum

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Student Workbook

Student Workbook — Module 2 Pañcabhūta (Senior)

Name: _____ Class: ____ Section: ____

Day 1 — A Persistent Question

Three frameworks I now know:

Framework	Date	# Elements
Greek (Empedocles, Aristotle)	5th–4th c. BCE	4 + 1
Sāṅkhya	pre-5th c. BCE → 4th c. CE	25 tattvas (5 are bhūtas)
Modern periodic table	1869 →	118+ elements

My question from SB 2.10.3: _____

Day 2 — The Tattva Ladder

(Draw the 25 tattvas here. Mark the five bhūtas distinctly.)

Day 3 — Quality-Count Rule

bhūta	tanmātra	indriya	# qualities
ākāśa	śabda	śrotra (ear)	1
vāyu	sparsa	tvac (skin)	2
tejas	rūpa	cakṣus (eye)	3

bhūta	tanmātra	indriya	# qualities
ap	rasa	rasanā (tongue)	4
pr̥thvī	gandha	ghrāṇa (nose)	5

SB 3.26.36 verse (write it here from memory or copy):

Day 4 — Greek Comparison

Dimension	Sāṅkhya	Greek
# elements	_____	_____
Origin	_____	_____
What drives combination	_____	_____
Includes ākāśa-like?	_____	_____
Perceptual mapping?	_____	_____

Day 5 — Primary Source Reading

(Paste the SB 3.26.36 handout here.)

My word-by-word reading went: [easy / hard / surprising]

What surprised me:

Day 6 — Modern Lens

Question	Modern	Sāṅkhya
Predicts chemistry	Strong	Weak
Categories of everyday world	OK	Strong
Perception & reality	Bracketed (chem)	Strong (tanmātra)
Consciousness	Bracketed	Central (puruṣa)

My quick-write position:

Day 7 — Debate

My position: _____

My 2 best arguments: 1. _____ 2. _____

Counter-argument I steel-manned: _____

Days 8–9 — Paper

Working title: _____

Thesis paragraph: _____

3 sources I'll cite: 1. _____ 2. _____ 3. _____

Strongest counter-argument from Day 9 buddy session: _____

Day 10 — Reflection

1. What did you set out to argue, and what did you actually argue? _____

2. Strongest counter-argument encountered? _____

3. With 3 more days, what would you research deeper? _____

Glossary

Sanskrit	Meaning
tattva	principle, “thatness”
prakṛti	primordial nature
puruṣa	consciousness, witness
guṇa	quality (Sāṅkhya: sattva, rajas, tamas)

Sanskrit	Meaning
sarga	manifestation, creation
avyakta / vyakta	unmanifest / manifest
bhūta / mahābhūta	gross element
tanmātra	subtle perceptual essence
indriya	sense organ
ākāśa	space / ether
śabda, sparśa, rūpa, rasa, gandha	sound, touch, form, taste, smell

Day 1 — Hook: A Persistent Question

Module: Pañcabhūta (Senior) · **Time:** 45 min

Learning objective

Students articulate what “categorising matter” means and identify three distinct historical frameworks for doing it.

Materials

- Whiteboard
- Periodic table poster
- Printed copy of SB 2.10.3 with IAST + translation (from cache)

Flow

Warm-up (5 min)

Opening provocation on board: “*Categories are not in nature. Categories are imposed on nature. Discuss.*”

Take 3 student responses.

Core (25 min)

1. **Three frameworks, one question.** Write on board:

Framework	Approximate date	Categories
Greek (Empedocles, Aristotle)	5th c. BCE → 4th c. BCE	4 + 1 (quintessence)
Sāṅkhya (India)	likely pre-5th c. BCE, codified c. 4th c. CE	25 tattvas (5 are bhūtas)
Modern periodic table	1869 (Mendeleev) → present	118+ elements

2. **The persistent question:** “*What categories of stuff (and not-stuff) does the world come in?*” Every culture’s natural philosophers tried to answer this. The answers differ because the methods differ and the questions are framed differently.

3. **Today’s anchor verse — SB 2.10.3.** Read in IAST. Then translation: > “*bhūta-mātreṇḍṛya-dhiyāṁ janma sarga udāhṛtaḥ ...*” > “The manifestation of the gross elements, the senses, and the intellect is called creation (*sarga*).”

Note the move: the Sāṅkhya verse names the bhūtas (elements), the indriyas (senses), and the dhī (intellect) as products of a single creation event. Modern science talks about atoms; Sāṅkhya talks about a unified scheme that includes the *perceiver*. This is a real structural difference, not just terminology.

Discussion (10 min)

Open question: “*Is the modern periodic table ‘more correct’ than Empedocles’ four elements? More correct in what way?*”

- Let students argue.
- Reinforce: a framework can be more *predictive* (chemistry, atomic structure) without being more *comprehensive* (the Sāṅkhya scheme includes consciousness; modern chemistry doesn’t claim to).

Wrap-up (5 min)

- Preview Day 2: “*Tomorrow we map out the full 25-tattva ladder of Sāṅkhya. The five elements are only the LAST five.*”
- Homework: read the SB 2.10.3 cache entry in full. Note one question you have.

Differentiation

- **Tier up:** Read Kirk, Raven & Schofield’s intro on Empedocles. Be ready to present on Day 4.
- **Tier down:** Stick to the framework-comparison; the primary-source reading deepens on Day 5.

Teacher notes

- Don’t let “modern science is just objectively right” be the resting position. Push: “what does objectively right *mean*? Predictive accuracy? Comprehensiveness? Explanatory power?”
- This module is about *intellectual honesty across frameworks*, not about advocacy for any one of them.

Citation(s) used

- SB 2.10.3 — vidya-karana RAG cache day-01-hook.json#chunks[0]

Day 2 — The Sāṅkhya Tattva Ladder

Module: Pañcabhūta (Senior) · **Time:** 45 min

Learning objective

Students draw and label all 25 tattvas in the Sāṅkhya scheme, and locate the five bhūtas within the full enumeration.

Materials

- Large whiteboard or pinboard
- A4 sheets for students
- Sources excerpt: Larson's *Classical Sāṅkhya* summary (teacher-prepared, 1 page)

Flow

Warm-up (5 min)

- 2 students share their homework question from Day 1.
- Quick recap: "Five elements emerge LAST in Sāṅkhya. Today we see what comes first."

Core (25 min)

1. Build the ladder on the board, slowly:

25. puruṣa (consciousness, the witness)
24. prakṛti (primordial nature, unmanifest)
_____ (evolution begins)
23. buddhi / mahat (intellect)
22. ahaṅkāra (ego, sense of "I")
_____ (branch into three lines)

Mental:	Sense organs:	Action organs:
21. manas (mind) (speech)	20. śrotra (ear)	15. vāk
(hand)	19. tvac (skin)	14. pāṇi
(foot)	18. cakṣus (eye)	13. pāda
(anus)	17. rasanā (tongue)	12. pāyu
(genitals)	16. ghrāṇa (nose)	11. upastha

Tanmātras (subtle elements, 10–6):
10. śabda-tanmātra (sound)
9. sparśa-tanmātra (touch)
8. rūpa-tanmātra (form)
7. rasa-tanmātra (taste)
6. gandha-tanmātra (smell)

Mahābhūtas (gross elements, 5–1):
5. ākāśa (space)
4. vāyu (air)

3. tejas (fire)
2. ap (water)
1. pṛthvī (earth)

(Numbering convention varies in different Sāṅkhya texts; here we count from the most concrete (1 = earth) up to the most abstract (24 = prakṛti) with puruṣa as the silent 25th. Other texts count from puruṣa down.)

2. The key structural insight:

- The five gross elements (*mahābhūtas*) are the LAST and most concrete layer.
- Each is paired with a *tanmātra* (subtle quality, the perceptual essence).
- Each *tanmātra* is what the corresponding sense-organ perceives.
- The whole scheme is internally consistent: the bhūtas, tanmātras, and indriyas line up in a tight one-to-one mapping.

Diagram exercise (10 min)

Students draw the 25-tattva ladder in their notebooks. Use colour to mark the five bhūtas distinctly.

Wrap-up (5 min)

- Quick check: “Where do the five elements sit in the ladder?” (At the bottom — the most concrete, most manifest layer.)
- Preview Day 3: “Tomorrow — the *bhūta-tanmātra* pairing in detail, and why the quality-count rule follows.”

Differentiation

- **Tier up:** Read the full first 22 verses of the Sāṅkhya Kārikā in translation. Note one verse you find puzzling.
- **Tier down:** Focus on the bhūtas + tanmātras layer. The full ladder can be a reference.

Teacher notes

- The 25-tattva count is the *classical Sāṅkhya* count (Īśvarakṛṣṇa’s *Sāṅkhya Kārikā*, c. 4th c. CE). Different versions of Sāṅkhya count differently.
- Be honest about the *non-theistic* character of classical Sāṅkhya. It does NOT include a creator god as a tattva. The Yoga school (*Yoga Sūtras*) adds *Īśvara* as a separate principle; classical Sāṅkhya does not.

Citation(s) used

- SB 3.5.36 — vidya-karana RAG cache day-02-creation.json#chunks[1]
- Larson, *Classical Sāṅkhya* (1979), chapters on tattva enumeration.

Day 3 — Quality-Count Rule and the Tanmātras

Module: Pañcabhūta (Senior) · **Time:** 45 min

Learning objective

Students explain the *bhūta–tanmātra* pairing and prove the quality-count rule (1, 2, 3, 4, 5 for sky, air, fire, water, earth) is internally consistent given the emergence sequence.

Materials

- Whiteboard
- Printed SB 3.26.36 + 3.26.44 (IAST + translation, from cache)

Flow

Warm-up (5 min)

- Quick recall: enumeration of the five bhūtas with their tanmātras.

Core (25 min)

1. **The quality-count rule** (write boxed):

Each element inherits all the qualities of the elements that emerged before it, AND adds one new quality of its own — the *tanmātra* associated with that element.

Element	Qualities	Source-verse anchor
ākāśa	śabda	SB 3.26 (qualities-of-space)
vāyu	śabda + sparśa	SB 3.26.36 (directly quoted: “softness, hardness, cold, heat — qualities of touch, tanmātra of air”)
tejas	śabda + sparśa + rūpa	—
ap	śabda + sparśa + rūpa + rasa	—
pṛthvī	śabda + sparśa + rūpa + rasa + gandha	SB 3.26.44 (taste → smell as the last new quality)

2. **Internal consistency:** the scheme is *symmetric and complete*:

- 5 bhūtas, 5 tanmātras, 5 indriyas of perception.
- Each bhūta pairs with exactly one tanmātra.
- Each tanmātra is perceived by exactly one indriya.
- The quality count (1, 2, 3, 4, 5) is a *predicted consequence* of the emergence order — not an arbitrary stipulation.

3. **Read SB 3.26.36 together** (in IAST): > “*mṛdutvaṃ kaṭhinatvaṃ ca śaityam uṣṇatvaṃ eva ca / etat sparśasya sparśatvaṃ tanmātratvaṃ nabhasvataḥ*” > “Softness, hardness, cold, and heat — these are the qualities of touch (*sparśa*), the subtle attribute (*tanmātra*) of air (*nabhasvat*).”

- Word-by-word with class: *mṛdutvam* = softness, *kaṭhīnatvam* = hardness, *śāityam* = cold, *uṣṇatvam* = heat. *etat* = these. *sparsāsyā sparśatvam* = the touch-ness of touch. *tanmātratvam nabhasvataḥ* = the subtle-essence-ness of air.
- Discussion: notice that *sparsā* (touch) here includes thermal sensations (cold, heat). This is broader than modern English “touch” (which usually means pressure/texture). The framework is more inclusive than we’d assume from a casual reading.

Concept check (10 min)

Worksheet question (1 mark each): - Why does *ākāśa* have only one quality? - Why does *prthvī* have five? - What does *sparsā* include that modern English “touch” doesn’t? - Why is the bhūta-tanmātra mapping called “one-to-one”? - If a sixth element existed, how many qualities would it have, and what would they be? (Discussion question.)

Wrap-up (5 min)

- Preview Day 4: “Tomorrow — Empedocles and Aristotle. Greek and Indian elemental schemes side by side.”

Differentiation

- **Tier up:** Write a 1-page response to “if a sixth element existed (e.g. ‘information’), where would it sit and what would its tanmātra be?”
- **Tier down:** Focus on the quality-count rule; the bhūta-tanmātra pairing alone is the core skill.

Teacher notes

- The *internal consistency* point matters more than students realise. Many will assume “old framework = primitive.” The Sāṅkhya enumeration is *not* primitive — it’s tightly self-consistent in a way that the Greek four-element scheme isn’t.
- The Empedocles scheme (Day 4) has NO equivalent of the tanmātra-indriya symmetry. That asymmetry is one of the most interesting cross-cultural points in the module.

Citation(s) used

- SB 3.26.36 — vidya-karana RAG cache day-03-qualities.json#chunks[3]
- SB 3.26.44 — vidya-karana RAG cache day-03-qualities.json#chunks[0]

Day 4 — Empedocles, Aristotle, and the Greek Four

Module: Pañcabhūta (Senior) · **Time:** 45 min

Learning objective

Students compare the Greek (Empedocles + Aristotle) four/five element scheme with Sāṅkhya, identifying at least three structural differences and one structural similarity.

Materials

- Printed extracts:
 - Empedocles fragments on the four roots (Kirk, Raven & Schofield translation)
 - Aristotle, *On Generation and Corruption*, II.3 (on the four elements + aither)
- Whiteboard
- Side-by-side comparison worksheet (blank)

Flow

Warm-up (5 min)

- Recall: the 5-bhūta + 5-tanmātra + 5-indriya structure of Sāṅkhya.

Core (25 min)

1. **Empedocles in brief.** Read 1-paragraph paraphrase aloud: > Empedocles (c. 490–430 BCE) proposed four “roots” (*rhizomata*) — earth, water, air, fire. He added two forces — Love (*philia*) that brings them together and Strife (*neikos*) that separates them. The world is the cyclic combination/separation of the four under these forces.
2. **Aristotle in brief:** > Aristotle (4th c. BCE) inherited the four sublunary elements (earth, water, air, fire) but added a *fifth element* — *aither* (the “quintessence”) — which is the substance of the heavenly spheres. The four are characterised by pairs of qualities: hot/cold and wet/dry.

Element	Aristotle’s qualities
Fire	hot + dry
Air	hot + wet
Water	cold + wet
Earth	cold + dry
Aither	none (incorruptible)

3. **Side-by-side comparison.** Build with class on the board:

	Sāṅkhya (India)	Empedocles + Aristotle (Greece)
Numerical number of elements	5	4 + 1 (quintessence)
Origin	Emerge sequentially from <i>prakṛti</i>	Eternal “roots”
World	Three <i>guṇas</i> (sattva, rajas, tamas)	Love and Strife (Empedocles); the natural motions toward

	Sāṅkhya (India)	Empedocles + Aristotle (Greece)
hat dri ve s co mb ina tio n	tamas)	“proper place” (Aristotle)
Fif th ele me nt ide nti ty	<i>Ākāśa</i> — universal medium of space, present everywhere	<i>Aither</i> — celestial substance, only in the heavens
Qu alit y- co unt rul e	1, 2, 3, 4, 5 (predicted from emergence order)	No equivalent — each element gets a pair of qualities (hot/cold + wet/dry)
Pe rce ptu al ma ppi ng (ta nm ātr a- ind riy a)	Yes — explicit, symmetric, one-to-one	No equivalent
Inc lud es co ns cio	Yes (puruṣa, 25th tattva)	No

Sāṅkhya (India)	Empedocles + Aristotle (Greece)
us	
ne	
ss	
in	
sc	
he	
me	

4. **Discuss the most interesting difference.** Open: “Which of these structural differences feels most significant to you? Why?”

Activity (10 min)

Begin in-class comparison essay draft: - Pick ONE structural difference from the table. - Write a 1-paragraph (5–7 sentence) analysis of why it matters. - This is a draft for the longer Day 8–10 paper.

Wrap-up (5 min)

- Preview Day 5: “Tomorrow — we read SB 3.26.36 closely in IAST, word by word.”

Differentiation

- Tier up:** Read Empedocles fragment B17 (on Love and Strife) in translation; write 1 paragraph on whether the *guṇa* doctrine in Sāṅkhya is structurally parallel.
- Tier down:** Focus on Aristotle alone; skip Empedocles for now.

Teacher notes

- The asymmetry on perceptual mapping (Sāṅkhya YES, Greek NO) is the most distinctive cross-cultural point. Push students to notice it.
- Some students will be tempted to conclude “Sāṅkhya is more sophisticated.” Push back: *more sophisticated in what way?* (More internal symmetry, yes. More predictive of chemical behaviour, no.)

Citation(s) used

- Kirk, G. S., Raven, J. E., & Schofield, M. (1983). *The Presocratic Philosophers* (2nd ed.). Cambridge University Press. — Empedocles fragments.
- Aristotle, *On Generation and Corruption*, II.3. (Any standard translation.)

Day 5 — Primary Source Reading + Mid-Module Check

Module: Pañcabhūta (Senior) · **Time:** 45 min

Learning objective

Students read SB 3.26.36 in IAST with word-by-word gloss and submit the formative quiz.

Materials

- Printed handout: SB 3.26.36 in Devanāgarī, IAST, word-by-word gloss, continuous translation
- Formative quiz (quizzes/formative.md Part B)

Flow

Warm-up (5 min)

- Hand out the printed verse. Quick recall: yesterday's Greek comparison.

Reading (15 min)

- First read — sound.** Teacher reads the IAST aloud once. Students listen.

*mṛdutvaṃ kaṭhinatvaṃ ca śaityaṃ uṣṇatvaṃ eva ca etat sparśasya sparśatvaṃ
tanmātratvaṃ nabhasvataḥ*

- Second read — meaning.** Students follow the word-by-word gloss:

Sanskrit	Gloss
mṛdutvaṃ	softness (nominative neuter)
kaṭhinatvaṃ	hardness
ca	and
śaityaṃ	cold
uṣṇatvaṃ	heat
eva ca	as well
etat	these
sparśasya	of touch (genitive)
sparśatvaṃ	the touch-ness (= the defining quality of touch)
tanmātratvaṃ	the subtle-essence-ness
nabhasvataḥ	of air / wind (genitive; nabhasvat = wind)

Continuous translation: > “Softness, hardness, cold and heat — these are the touch-ness of touch, the subtle essence of air.”

- Class discussion (8 min):**

- What surprises you about the categories included under “touch”?
- Note the philosophical move: *sparśasya sparśatvaṃ* (touch-ness of touch) — the verse identifies the defining property of a category by giving its essence a name.
- How is this different from a modern dictionary definition of “touch”?

Formative quiz (15 min)

- quizzes/formative.md Part B.

Wrap-up (5 min)

- Preview Day 6: “*Tomorrow — the modern lens. What modern science adds, and what it explicitly drops.*”

Differentiation

- **Tier up:** Write a 1-paragraph response to “what is gained when philosophy gives an *essence* a name?”
- **Tier down:** Focus on the meaning. The IAST pronunciation is bonus.

Teacher notes

- Don’t require pronunciation perfection. The win is *engagement with the meaning at primary-source level*.
- This is many students’ first time reading a Sanskrit verse. Honour the moment.

Citation(s) used

- SB 3.26.36 — `vidya-karana RAG cache day-03-qualities.json#chunks[3]`

Day 6 — The Modern Lens

Module: Pañcabhūta (Senior) · **Time:** 45 min

Learning objective

Students identify three things modern science *adds* to the picture that Sāṅkhya does not have, and three things Sāṅkhya *includes* that modern science explicitly excludes.

Materials

- Periodic table poster
- NCERT Class 11 Chemistry, Chapter 1 (atomic structure) — printed extract
- Whiteboard

Flow

Warm-up (5 min)

- Hand back yesterday’s quiz. Brief review of common errors.

Core (25 min)

1. **What modern science adds:**
 - **Atomic structure** — protons, neutrons, electrons. The periodic table’s 118 elements are atom-based, not state-of-matter-based.
 - **Predictive chemistry** — combinations follow rules (valence, electronegativity) we can calculate.

- **Subatomic and field-level descriptions** — quarks, electromagnetic field, weak and strong forces. None of this is in Sāṅkhya.
 - **Quantitative measurement** — mass, density, melting point. Sāṅkhya is qualitative.
2. **What Sāṅkhya includes that modern science explicitly excludes:**
- **Empty space (*ākāśa*) as a category** — modern physics treats vacuum as the absence of matter, not as a distinct category.
 - **A perceptual mapping** — *tanmātra-indriya* one-to-one symmetry. Modern science treats perception as biology (neuroscience), not as a metaphysical layer.
 - **Consciousness in the scheme** — *puruṣa* (the witness) is part of the 25-tattva enumeration. Modern science either bracket consciousness (default in chemistry) or treats it as emergent from matter (in some neuroscience interpretations).
3. **The honest framing:** these are different KINDS of inquiry, not different *qualities* of inquiry.

Inquiry	Modern science	Sāṅkhya
“What predicts chemical behaviour?”	Strong	Weak
“What categories does the everyday world come in?”	Decent (states of matter)	Strong
“How does perception fit with reality?”	Modern: neuroscience; not in chemistry	Strong (<i>tanmātra-indriya</i>)
“Where does consciousness fit?”	Mostly bracketed	Central (<i>puruṣa</i>)

Reflection writing (10 min)

5-minute quick-write: “For one question above where Sāṅkhya is ‘Strong’ but modern science is ‘Mostly bracketed’ or ‘Weak,’ do you think this is a real intellectual asset of the Sāṅkhya scheme, or a leftover of pre-scientific guessing? Defend your position in 4–5 sentences.”

Discussion (5 min)

3 students share their position. Note the diversity of views.

Wrap-up (5 min)

- Preview Day 7: “Tomorrow — group debate. Should IKS be integrated into mainstream science class?”

Differentiation

- **Tier up:** Read Thomas Nagel’s “What Is It Like to Be a Bat?” — write 1 paragraph on whether Sāṅkhya’s *tanmātra-indriya* symmetry maps onto Nagel’s “subjective experience” question.

- **Tier down:** Stick to the comparison table; the quick-write can be 2–3 sentences.

Teacher notes

- Push students past “old vs new” framing into “different scopes, different methods.”
- Some students may genuinely believe consciousness is “just neurons” — that’s a defensible position. Some may believe it’s irreducible. Both are taken seriously in contemporary philosophy of mind; the module shouldn’t decide for them.

Citation(s) used

- NCERT Class 11 Chemistry, Chapter 1.
- (Optional) Nagel, T. (1974). “What Is It Like to Be a Bat?” *Philosophical Review*, 83(4), 435–450.

Day 7 — Discussion: Should IKS Be in Science Class?

Module: Pañcabhūta (Senior) · **Time:** 45 min

Learning objective

Students articulate and defend a position on whether classical Indian intellectual frameworks should be taught in mainstream science class, with at least one citation and one steel-manned counter-argument.

Materials

- Whiteboard
- Stopwatch
- Score sheet for the oral assessment (rubric below)

Flow

Warm-up (3 min)

- Brief: today is structured debate. Three positions on the table. Pick one. Defend it. Acknowledge the other two fairly.

The three positions (5 min — clarify, don’t argue yet)

Write on board:

Position	One-line claim
A — Integration	IKS frameworks should be taught alongside modern science in mainstream science class.
B — Separation	IKS should be taught in cultural studies or philosophy, not in science class.
C — Hybrid	IKS should be referenced briefly in science class as historical context, but not taught as parallel content.

Debate (25 min)

- Divide class into 3 groups by position (let students self-select; balance roughly).
- 5 min: each group prepares — strongest 2 arguments, 1 steel-manned response to the other two positions.
- 12 min: structured debate. Each group has 4 min.
 - 2 min: main argument.
 - 1 min: response to the other 2 positions.
 - 1 min: closing.
- 8 min: open floor cross-questions.

Oral assessment (during the debate)

Teacher uses this rubric to score 5 students chosen at random:

Criterion	4	3	2	1
Position clarity	Crystal clear with 1-line summary	Clear	Mostly clear	Unclear
Use of evidence	Cites at least 1 specific source	Cites generally	Vague references	No evidence
Counter-argument	Steel-mans an opposing view	Acknowledges another view	Mentions another view	Ignores
Listening	Responds to specific points raised	Responds generally	Sticks to script	Doesn't engage

Wrap-up (10 min)

- Take 3 student reflections: “*Did anyone’s mind change during the debate?*”
- Note: the goal is not to “win” but to *understand the strongest version of each position*.
- Preview Day 8: “*Tomorrow you start your paper. Use this debate to sharpen your own thesis.*”

Differentiation

- **Tier up:** Take a position you DISAGREE with and defend it. (The hardest form of argument.)
- **Tier down:** Pre-prepare your 2-minute argument the night before.

Teacher notes

- This is where some students discover they don't really know what they think. That's the point.
- If the debate gets heated, slow it down: "*Steel-man the other side. What's their best argument?*"
- The Day 7 oral grade flows into the project rubric — not as a separate score.

Citation(s) used

- All sources from `sources.md` are fair game during the debate.

Day 8 — Project Session 1: Outline Your Paper

Module: Pañcabhūta (Senior) · **Time:** 45 min

Learning objective

Students choose a focused paper topic, draft a one-paragraph thesis, and produce a working outline.

Materials

- Project brief (`assessments/project-brief.md`)
- Rubric (`assessments/rubric.md`)
- Laptop or notebook for drafting

Flow

Warm-up (5 min)

- Hand out the rubric. Read aloud the *thesis* and *counter-argument* criteria.

Core: outlining (30 min)

Sprint 1 (10 min) — Topic. Each student writes: - Working title (1 line) - One-paragraph thesis (3–4 sentences) - 3 sources they'll cite (must include one primary scriptural, one secondary scholarly, one comparative or modern)

Teacher rotates: 60 sec per student, single question: "*What's the counter-argument someone could make against your thesis?*"

Sprint 2 (10 min) — Outline. Each student writes section headings + 1-line sub-points: - Introduction (with thesis statement) - Section 1 (anchor claim + source) - Section 2 (anchor claim + source) - Counter-argument and response - Conclusion

Sprint 3 (10 min) — Begin drafting. Each student starts the introduction or one body paragraph.

Wrap-up (10 min)

- Each student exchanges outline with one buddy.
- 3-min mutual feedback: 1 strength + 1 fix.
- Outline + thesis submitted at the door.

Homework

- Draft 1–2 more paragraphs at home. Bring everything tomorrow.

Differentiation

- **Tier up:** Add a 4th source — a contemporary academic article (search Google Scholar).
- **Tier down:** Topic can be narrow (“a comparison of one Sāṅkhya tanmātra concept with the modern five-senses model”). Narrower is fine.

Teacher notes

- Common pitfall: too-broad topic. Push for specificity early.
- A weak thesis sounds like “Indian science is interesting.” A strong thesis sounds like “The Sāṅkhya tanmātra–indriya symmetry offers a model that 19th-century European psychophysics independently rediscovered.”

Day 9 — Project Session 2: Draft + Counter-Arguments

Module: Pañcabhūta (Senior) · **Time:** 45 min

Learning objective

Each student reaches at least 2 drafted pages including a fully-articulated counter-argument and response.

Materials

- Drafts in progress
- Rubric

Flow

Warm-up (5 min)

- Brief: today’s focus is the counter-argument. The rubric rewards a *fair* presentation of the strongest opposing view.

Core (30 min)

Sprint 1 (15 min) — Draft body. Quiet writing. Aim for 1.5–2 pages drafted.

Sprint 2 (15 min) — Counter-argument workshop. - Pair with a buddy. - Read each other’s drafts so far. - Each pair argues: “*Take my thesis. Give me the strongest counter-argument you*

can think of.” - The student takes notes on the counter-argument and decides how to respond in their paper.

Wrap-up (10 min)

- 3 students share their best-discovered counter-argument.
- Outline-to-date submitted at door (revised since Day 8).

Homework

- Finish a complete draft at home tonight. Bring tomorrow.

Differentiation

- **Tier up:** Construct TWO counter-arguments — one philosophical, one practical.
- **Tier down:** Submit a clear outline + 1 paragraph if a full draft is too much.

Teacher notes

- Some students will skip the counter-argument because it feels like “weakening” their position. Reframe: a strong paper anticipates and answers objections.

Day 10 — Summative Quiz + Paper Submission

Module: Pañcabhūta (Senior) · **Time:** 45 min (paper accepted up to end of day)

Learning objective

Students complete the summative quiz and submit their 5-page argumentative paper.

Materials

- Summative quiz (quizzes/summative.md)
- Stopwatch

Flow

Quiz (35 min)

- Distribute. Closed-book.
- 35 marks. Multiple choice, short answer, primary-source reading question, comparison question, open response.

Paper submission (10 min)

- Hand in printed papers (or upload digitally).
- Teacher quick-checks each for:
 - Title page with name + thesis
 - Page count (4.5–5.5 pages target)
 - Works-cited page with at least 3 sources

- Any paper missing a piece is flagged for late penalty.

Wrap-up

- Brief class closure.
- Reflection slip submitted (3 questions from `assessments/project-brief.md`).

Homework

- None. Module complete. (Paper marked within 1 week.)

Differentiation

- **Tier up:** Students who submit early can begin reading for the next module's prerequisites.
- **Tier down:** Open-notebook quiz for documented learning needs.

Teacher notes

- Mark papers in two passes: first pass for clarity and structure; second pass for source quality and counter-argument fairness.
- Return marked papers within 5 school days with detailed comments.

Citation(s) used

- All sources verified.

Writing Prompts

Writing Prompts — Module 2 Pañcabhūta (Senior)

Eight prompts at essay-level depth.

Reflection prompts

- (Day 1)** Categories are conventions imposed by minds, not features of nature. Defend or refute in 1 page, using examples from the periodic table and the Sāṅkhya tattva enumeration.
- (Day 6)** Modern science excludes consciousness from chemistry. Sāṅkhya includes puruṣa as the 25th tattva. Is this an asset or a liability of the Sāṅkhya scheme? 1 page.

Explanatory prompts

- (Day 3 or Day 5)** In 1 page, explain to a college-level reader why the quality-count rule (1, 2, 3, 4, 5) is a *predicted consequence* of the Sāṅkhya emergence sequence, not an arbitrary stipulation. Use SB 3.26.36 as a primary source.
- (Day 4)** Compare and contrast the role of Empedocles's Love-and-Strife with the role of the three guṇas (sattva, rajas, tamas) in Sāṅkhya. 1 page.

Argumentative prompts

5. (Day 7 or Day 8 — could be paper topic) Argue for or against the claim: “Sāṅkhya’s bhūta-tanmātra-indriya symmetry is structurally more complete than Aristotle’s element-quality scheme.” 2 pages, with sources.

6. (Senior research) Argue: “The Sāṅkhya scheme’s inclusion of ākāśa as a positive category (rather than as absence-of-matter) anticipates a question that modern field theory addresses but did not solve until the 20th century.” 2 pages.

Cross-disciplinary prompts

7. (Day 6) Pick a contemporary debate (consciousness studies, panpsychism, the hard problem of consciousness) and write 1 page on how Sāṅkhya’s tattva enumeration speaks to it — or fails to.

8. (Stretch) Write a 2-page comparison of the *guṇa* doctrine in Sāṅkhya with the *element* doctrine in classical Chinese philosophy (wǔxíng — Wood, Fire, Earth, Metal, Water). Use one specific source from each tradition.

How to grade essays

- Strong: clear thesis, specific citations, fair counter-argument, original thinking, clean prose.
- Adequate: clear thesis, some citations, weak counter-argument, mostly competent prose.
- Weak: vague thesis, no citations, no counter-argument, structural issues.

Homework

Homework — Module 2 Pañcabhūta (Senior)

Day 1 → Day 2

Reading: SB 2.10.3 from the cache (IAST + translation). Write 1 question you have about the verse.

Time: 20 min.

Day 2 → Day 3

Reading: Larson, *Classical Sāṅkhya*, opening chapter on tattva enumeration (teacher’s 1-page summary).

Time: 25 min.

Day 3 → Day 4

Reading: Empedocles fragment B17 (translation in handout). Note: how does Love-and-Strife differ from the Sāṅkhya guṇa doctrine?

Time: 25 min.

Day 4 → Day 5

Practise: Try reading SB 3.26.36 in IAST aloud (in handout). Don't worry about perfect pronunciation.

Time: 15 min.

Day 5 → Day 6

Writing: Submit the in-class quiz follow-up — 1 paragraph reflection on what reading a Sanskrit verse felt like.

Time: 20 min.

Day 6 → Day 7

Preparation for debate: Choose your position (Integration / Separation / Hybrid). Prepare your 2-minute opening argument + 1 steel-manned response.

Time: 30 min.

Day 7 → Day 8

Submit project plan: working title, thesis paragraph, 3 source citations.

Time: 30 min.

Day 8 → Day 9

Draft: Aim for 1.5–2 pages drafted at home.

Time: 60–90 min.

Day 9 → Day 10

Finalise: Complete draft, proofread, format. Submit tomorrow.

Time: 60–90 min.